User Manual

Constant Fraction Discriminator

CFD-X02-A



User Manual – Version v1.2.1
15 June 2026

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1 General Information

This chapter provides essential information on the purpose of this manual, the manufacturer, regulatory compliance, warranty conditions, and the conventions used throughout this document.

1.1 About This Manual

This manual provides instructions for the installation, configuration, operation, maintenance, and disposal of the Constant Fraction Discriminator. It forms an integral part of the product and must remain accessible to the personnel responsible for the device. If the device is transferred to a third party, all accompanying documentation must be delivered with it.

This manual contains proprietary information and may not be reproduced or distributed, in whole or in part, without prior written consent of the manufacturer.

Before performing any operation, the entire manual must be read carefully and the relevant sections reviewed in detail.

1.1.1 User Qualification

The device is intended for use by qualified personnel familiar with laboratory electronic instrumentation and responsible for the installation, operation, and maintenance of the device throughout its service life. It is not a toy and must not be used by children under 14 years of age. Persons with reduced physical, sensory, or cognitive abilities may operate the device only under supervision by qualified personnel.

1.1.2 Revision History

Document Revision: v1.2.1- 15 June 2026

A certified copy of this manual is stored in the manufacturer's technical file for the legally required period (10 years). During this period, documentation may be requested from the manufacturer.

1.2 Symbols and Conventions

The following graphical symbols and textual conventions may appear in this manual or on the device.

Provides important supplementary information.

Indicates procedures that, if not observed, may result in damage to the device or associated equipment.

Indicates procedures that, if not observed, may result in injury or serious damage.



(a) General hazard symbol



(b) Mandatory reading of the manual

Figure 1.1: The device may include the pictograms shown above.

All safety symbols must remain clearly visible throughout the lifetime of the device.

1.3 Manufacturer Information

Manufacturer: FLIM LABS S.r.l.
Address: Via della Farnesina, 3 00135 Rome (RM), Italy
Telephone: +39-329 761540
E-mail: info@flimlabs.com
Website: www.flimlabs.com

1.4 Declaration of Conformity

The device is supplied with a Declaration of Conformity drawn up in accordance with applicable European legislation. Before use, verify that the Declaration of Conformity is present. A copy of the Declaration of Conformity is shown in Document 1.1.

1.5 Product Label

The product label affixed to the device contains the required regulatory markings and identification information including:

- Product name and model
- Manufacturer identification
- Regulatory markings
- Electrical specifications

An example product label is shown in Figure 1.2. Disposal requirements associated with the WEEE marking are described in Section 7.2.

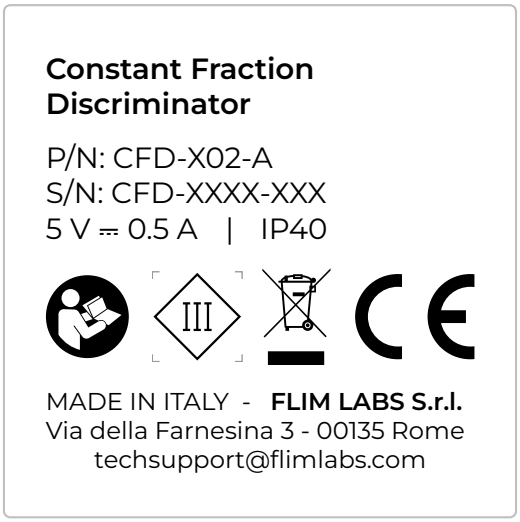


Figure 1.2: Product label

Safety pictograms performing a protective function must not be removed, covered, or damaged. If the label becomes illegible or damaged, contact authorized service for replacement.

1.6 Warranty and Liability

1.6.1 Warranty

The manufacturer provides 24 months warranty for private customers and 12 months warranty for commercial customers. The warranty period begins on the date of delivery.

EU Declaration of Conformity

CE

Device Information	
Type:	Constant Fraction Discriminator
Model:	Constant Fraction Discriminator
Part Number:	CFD-X02-A
Serial Number:	As indicated on the product label

Manufacturer Information	
Manufacturer:	FLIM LABS S.r.l.
Address:	Via della Farnesina, 3 00135 ROME (RM), Italy
Telephone:	+39 – 329 761540
E-mail:	info@flimlabs.com
Website:	www.flimlabs.com

The present declaration of conformity is issued under the sole responsibility of the manufacturer. The device complies with the following directives and regulations:

Applicable Directives and Regulations

- Directive 2014/35/EU, known as the “Low Voltage Directive”
- Directive 2014/30/EU, known as the “Electromagnetic Compatibility Directive”
- Directive 2011/65/EU, known as “RoHS”
- Delegated Directives (EU) 2015/863 and (EU) 2017/2102 amending Directive 2011/65/EU
- Directive 2012/19/EU, known as “WEEE”
- Regulation (EU) 2023/988 on General Product Safety

The manufacturer undertakes to provide, upon a duly reasoned request from the national authorities, all relevant information regarding the device.

Signed for and on behalf of FLIM LABS S.r.l.:

Place and Date

Rome, 24/10/2025

Signature

Alessandro Rossetta

Alessandro Rossetta, CEO

Document 1.1: EU Declaration of Conformity

The warranty covers defects in materials and workmanship under normal and intended use. It is valid only if the device is operated and maintained according to this manual. Unauthorized modification, improper use, failure to follow instructions, or repair by non-authorized personnel voids the warranty. Full warranty terms are available in the manufacturer's General Terms and Conditions published on the official website.

To submit a warranty claim, provide the following:

- Product type/version
- Proof of purchase (including date)
- Detailed description of the issue

1.6.2 Limitation of Liability

The device has been designed, manufactured, and tested in accordance with applicable European directives and internal quality standards. The manufacturer is liable exclusively for defects attributable to manufacturing faults or non-conformity with declared specifications, within the limits defined in the warranty terms.

The manufacturer shall not be liable for damages resulting from improper use, including but not limited to:

- Use outside the intended purpose
- Failure to follow this manual or applicable safety instructions
- Installation or operation by unqualified personnel
- Operation outside specified electrical or environmental limits
- Use of incompatible or non-compliant input signals or connected equipment
- Unauthorized modification, repair, or use of non-original spare parts
- Integration into systems not compliant with applicable safety regulations

The user is responsible for ensuring that the device is installed, operated, and integrated in accordance with this manual and all applicable local, national, and international regulations.

Except where otherwise required by mandatory law, the manufacturer shall not be liable for indirect, incidental, or consequential damages, including but not limited to loss of data, loss of profit, or system downtime.

2 Device Description

This section describes the functional characteristics, intended application, and technical features of the Constant Fraction Discriminator.

2.1 Functional Description

The Constant Fraction Discriminator (CFD-X02-A) is a compact, high-performance electronic module designed for precise timing discrimination of fast analog pulses in photon detection, nuclear electronics, spectroscopy, and time-resolved microscopy applications.

It converts analog input signals (e.g., from PMTs, APDs, or pulsed laser synchronization outputs) into well-defined digital timing signals. By applying the constant fraction discrimination principle, the device determines pulse arrival time independently of amplitude, significantly reducing timing walk.

The module features:

- Single-channel architecture
- Four simultaneous digital outputs
- Two SMA outputs (2.5 V logic, 50 Ω)
- Two USB Type-C LVDS outputs
- Selectable polarity switch
- Adjustable output pulse width (5 ns – 250 ns)
- Delay adjustment via external SMA coaxial delay cable
- Compact aluminum enclosure suitable for system integration
- USB Type-C power supply

2.2 Intended Use

The device is intended as a compact electronic module for converting analog to digital timing signals by professional users in research laboratories, universities, or industrial environments. It may be used as a stand-alone unit or integrated into custom measurement systems.

2.3 Reasonably Foreseeable Misuse

The following uses are not permitted:

- Use outside described applications
- Use with incompatible components
- Operation outside specified limits
- Unauthorized modification
- Use with incorrect power supply

2.4 Components Overview

Figure 2.1 shows an overview of the key components of the device. Those include:

1. Polarity selector
2. SMA input
3. Input threshold potentiometer
4. Test point
5. SMA monitor output
6. LED power indicator
7. Output pulse width potentiometer

8. Digital outputs 1 (SMA / USB-C LVDS)
9. Digital outputs 2 (SMA / USB-C LVDS)
10. Power input (USB-C)
11. Delay line connectors (SMA)
12. Grounding point



Figure 2.1: Front and back view showing the key components of the Constant Fraction Discriminator.

2.5 Mechanical Drawings and Dimensions

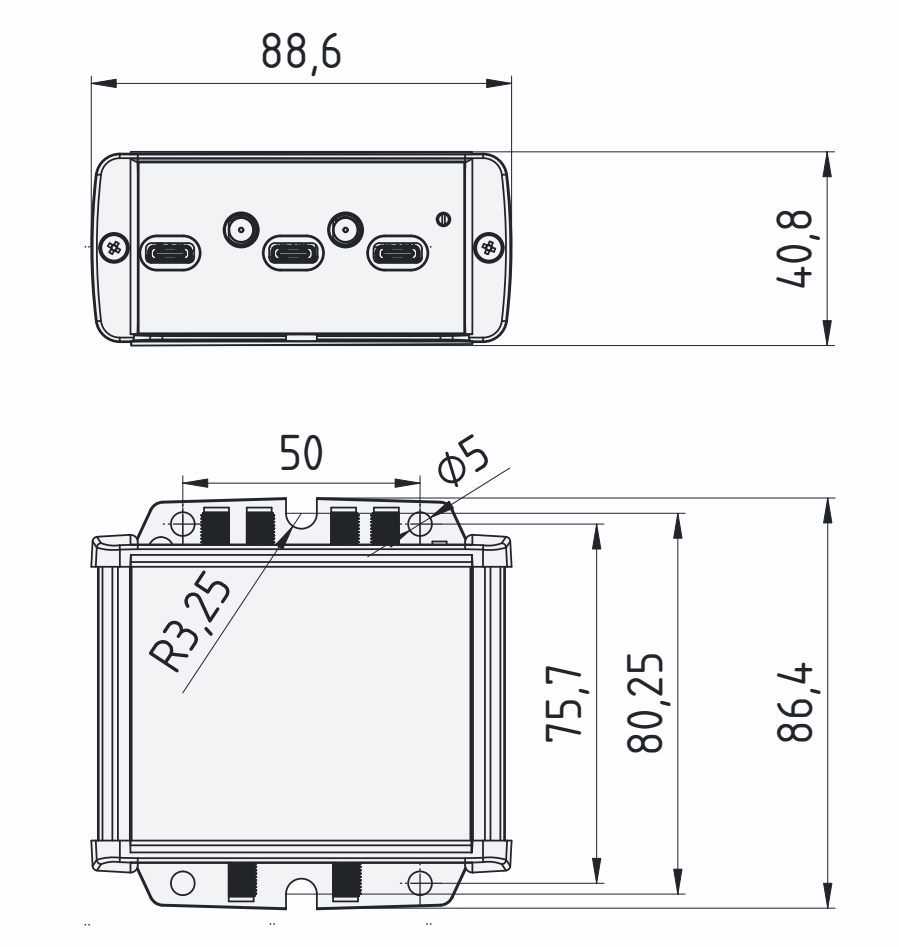


Figure 2.2: Technical drawing of the Constant Fraction Discriminator.

2.6 Technical Specifications

The electrical, signal, and environmental limits referenced throughout this manual are summarized in this section.

Item	CFD-X02-A
Electrical Specifications	
Electrical safety class	Class III
Power supply	5 V DC
Current consumption (max)	0.5 A
Power consumption (max)	2.5 W
Signal Specifications	
Input Connections	1x SMA
Input polarity	positive or negative
Input pulse width	>700 ps
Rise time	>350 ps
Minimum input pulse height (abs)	100 mV
Maximum input pulse height (abs)	5 V limited
Output pulse width	from 5 ns to 250 ns
Output connections and type	2x SMA: 2.5 V digital LVTTL at 50 Ω 2x USB Type-C: LVDS
Timing jitter	<15 ps
Time walk	±100 ps[a]
Maximum operating frequency	100 MHz
Length of delay line	adjustable based on external cable length (30 cm ~1ns)
Impedance	50 Ω
Physical Specifications	
Dimensions (L x W x H)	87 mm x 87 mm x 40 mm
Weight	225 g
Protection rating	IP40
Environmental Specifications	
Operating temperature	0 °Cto +40°C
Storage temperature	-20 °Cto +70°C
Humidity	80% non-condensing

[a] Measurement conditions: input pulse amplitude = -100 mV to -4.5 V, fall time = 1 ns, pulse width = 2.5 ns, CFD delay = 1 ns

Table 2.1: Technical specifications of the Constant Fraction Discriminator.

3 Safety Information

3.1 General Safety Instructions

Before installation and operation, ensure that this manual is complete and legible. If any part of the documentation is missing or unclear, contact the manufacturer before proceeding.

The device must be used only as described in this manual. Any operation not explicitly described in this manual is considered improper use.

This manual complements applicable national and local safety regulations. If local regulations impose stricter requirements, those regulations take precedence.

Despite the implemented safety measures, residual risks may remain if the device is not used as specified.

3.2 Electrical Safety

- Operate the device only with the specified power supply.
- Use undamaged, properly rated cables.
- Disconnect power before modifying signal connections.
- Do not open the enclosure. There are no user-serviceable internal components.
- Keep away from water sources and other liquids.
- Unauthorized repair or modification is not permitted.

3.3 Environmental and Operating Conditions

The device is designed for indoor laboratory use only. Operate the device only within the specified environmental limits:

- Operating temperature: 0°Cto +40°C
- Storage temperature: −20°Cto +70°C
- Relative humidity: up to 80% non-condensing
- Stable, vibration-free surface

Do not operate the device in explosive atmospheres or in the presence of flammable gases.

Ensure adequate ambient lighting during installation and adjustment to avoid connection errors.

3.4 Modification and Improper Use

The device must not be modified, altered, or repaired by unauthorized personnel. Use outside the intended application, operation beyond specified electrical limits, or failure to follow the procedures described in this manual may result in malfunction and void the warranty.

3.5 Documentation Retention

This manual must be kept accessible to personnel responsible for installation, operation, and maintenance. A replacement copy may be requested from the manufacturer if required.

4 Setup

This section describes the correct procedure for unpacking, installing, and configuring the device before operation.

4.1 Scope of Delivery

- CFD Module
- SMA coaxial cable (fixed delay line)
- USB Type-C power cable
- Threshold calibration screwdriver
- Quick start guide
- Declaration of conformity

Verify that all items are present before proceeding with installation.

4.2 Unpacking and Inspection

The device is supplied in protective packaging to protect it during storage and transport. Handle the device carefully during unpacking to avoid mechanical damage.
After removing the packaging:

- Inspect the device and accessories for visible damage.
- Check for dents, deformation, cracks, or signs of impact.
- Verify that connectors and cables are intact.

If any damage, missing parts, or abnormalities are detected, stop the installation and contact the manufacturer before proceeding.
Dispose of packaging materials responsibly and keep them out of reach of children.
The device is compact and lightweight and may be moved manually. When handling, support it securely to prevent accidental dropping.

4.3 Installation Requirements

Before installation and operation, ensure that the environmental conditions specified in Section 3.3 are met. Children must not have access to the installation or storage area. The device requires a regulated 5 V DC power source via USB Type-C capable of supplying at least 1 A.



Before connecting the device, verify that the input signal characteristics are compatible with the technical specifications of the CFD(see Table 2.1). Improper signal conditions may result in malfunction or unreliable operation. An overview of the required input signal checks is provided in Figure 4.1.

4.4 Mechanical Installation

Place the device on a stable laboratory surface with adequate space for cable routing and ventilation.
Ensure that:

- The device is positioned to prevent mechanical stress on connectors.
- Cables are routed safely and do not create tripping hazards.
- The power supply location is dry and away from water sources (e.g., sinks or splash zones).

Remove any protective dust or moisture covers before operation.

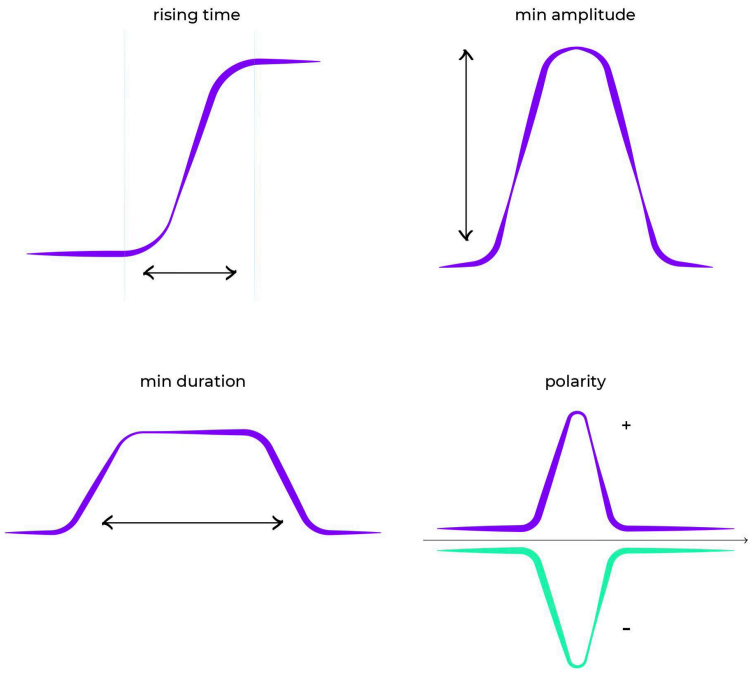


Figure 4.1: Verify that the input signal characteristics.

4.5 Electrical Connections

All electrical connections must be performed by qualified personnel who have read this manual. Connector locations are shown in Figure 2.1.

Proceed as follows:

1. **Power Connection**:
   a) Connect the USB Type-C cable to the device power input.
   b) Connect the other end to a suitable 5 V DC power source rated at a minimum of 1 A (e.g., computer USB port, certified USB power adapter, or power bank).
   c) Verify that the Power LED is illuminated.
2. **Delay Line Installation**:
   Install the appropriate external SMA delay cable between the delay output and delay input (see Figure 4.2).

> The delay length must match the rise time of the input signal. As a guideline, a 30 cm delay line is suitable for a signal rise time of approximately 1 ns. Proper delay selection is essential for optimal constant fraction discrimination.

3. **Input Signal Connection**:
   Connect the analog input signal to the SMA input port.
4. **Polarity Selection**:
   Set the polarity selector switch according to the polarity (positive or negative) of the input signal.
5. **Monitoring Connection**:
   Connect the monitor output and one digital output to an oscilloscope for adjustment and verification.
6. **Output Signal Connection**:
   Connect signal cables to the SMA or USB output ports as required for your timing setup.

> The CFD sends positive output signals. Only the rising edge of those signals provides reliable timing and should be used for reference. See Table 2.1 for signal specifications.

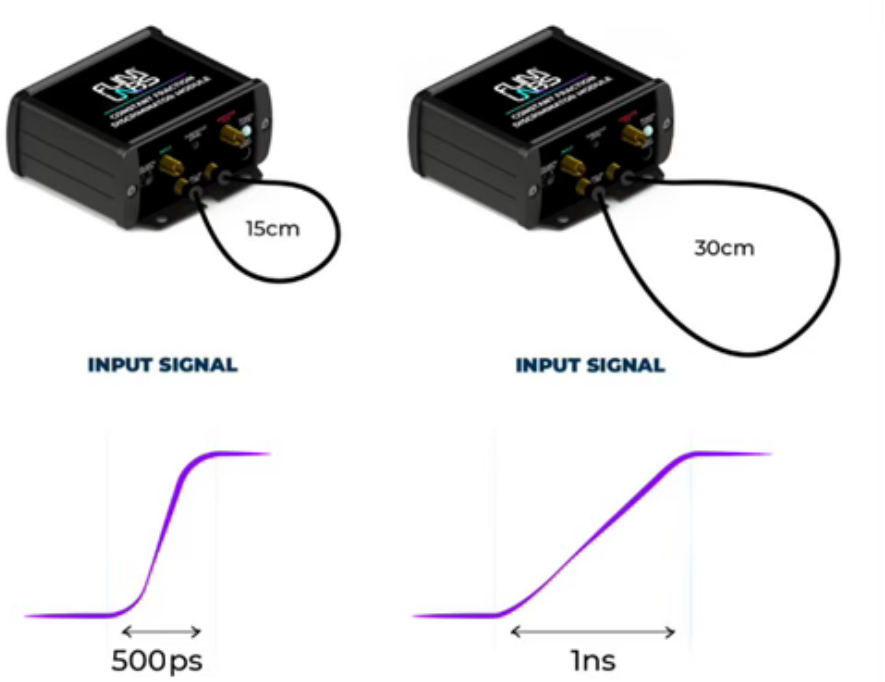


Figure 4.2: Matching of delay line cable to input signal.

4.6 Initial Configuration

After completing all electrical connections, two parameters need to be configured to ensure optimal performance of the CFD:

4.6.1 Input Threshold Adjustment

The input threshold defines the discrimination level of the analog input signal. Proper adjustment ensures reliable timing while avoiding triggering on noise.

1. Connect an oscilloscope to the monitor output and one of the digital outputs.
2. Adjust the input threshold potentiometer using the supplied screwdriver.
3. Gradually increase the threshold while observing the signals until:
   - The monitor signal crosses the zero reference cleanly.
   - A stable digital output pulse is generated.
   - Noise-induced false triggering is minimized.

4.6.2 Output Pulse Width Adjustment

The pulse width determines the duration of the generated digital output signal. Excessively long pulses may limit the maximum count rate of the system.

1. Using an oscilloscope, adjust the output pulse width potentiometer.
2. Set the pulse duration within the range of approximately 5 ns to 250 ns according to the requirements of the downstream electronics or data acquisition system.

Once these steps are completed, the device is ready for normal operation.

5 Operation

This section describes the functional behavior of the Constant Fraction Discriminatorduring use, including its operating principle, normal operating conditions, adjustable parameters, and troubleshooting guidance.

5.1 Principle of Operation

The CFD converts fast analog input pulses into precisely timed digital output signals using the constant fraction discrimination principle.

Unlike simple threshold discriminators, the CFD determines the arrival time of a pulse independently of its amplitude. By generating a fractionally scaled and delayed version of the input signal and detecting the zero-crossing point, the device significantly reduces timing walk caused by pulse height variations. The device provides:

- One analog input (SMA)
- Monitor output for signal observation
- Two digital SMA outputs (2.5 V LVTTL, 50 Ω)
- Two LVDS outputs via USB Type-C

The device does not perform signal amplification or conditioning beyond discrimination. Therefore, input signals must already be properly conditioned and compliant with the specified electrical limits. Typical applications include timing of photomultiplier tubes (PMTs), avalanche photodiodes (APDs), pulsed laser synchronization signals, and other fast analog pulse sources.

5.2 Normal Operation



Do not operate the device if it is damaged or shows signs of unsafe condition.

Once the power supply, input signal, and output connections have been correctly established and the initial configuration has been completed, the CFD operates autonomously without further user intervention. Correct functionality depends on proper system integration, and the user is responsible for ensuring that the complete measurement setup meets applicable safety and performance requirements. A typical example of the configured input and output waveforms is shown in Figure 5.1.

5.3 Parameter Adjustment

During operation, minor adjustments may be required to optimize timing performance or adapt to changes in input signal conditions.

The CFD provides two adjustable parameters:

Input Threshold: defines the discrimination level of the analog input signal.

Output Pulse Width: defines the duration of the digital output pulse.

Detailed step-by-step instructions for adjusting these parameters are provided in Section 4.6.

After adjustment, verify stable and consistent digital output behavior under expected operating conditions. Typical integration examples are shown in Figures 5.2 and 5.3.

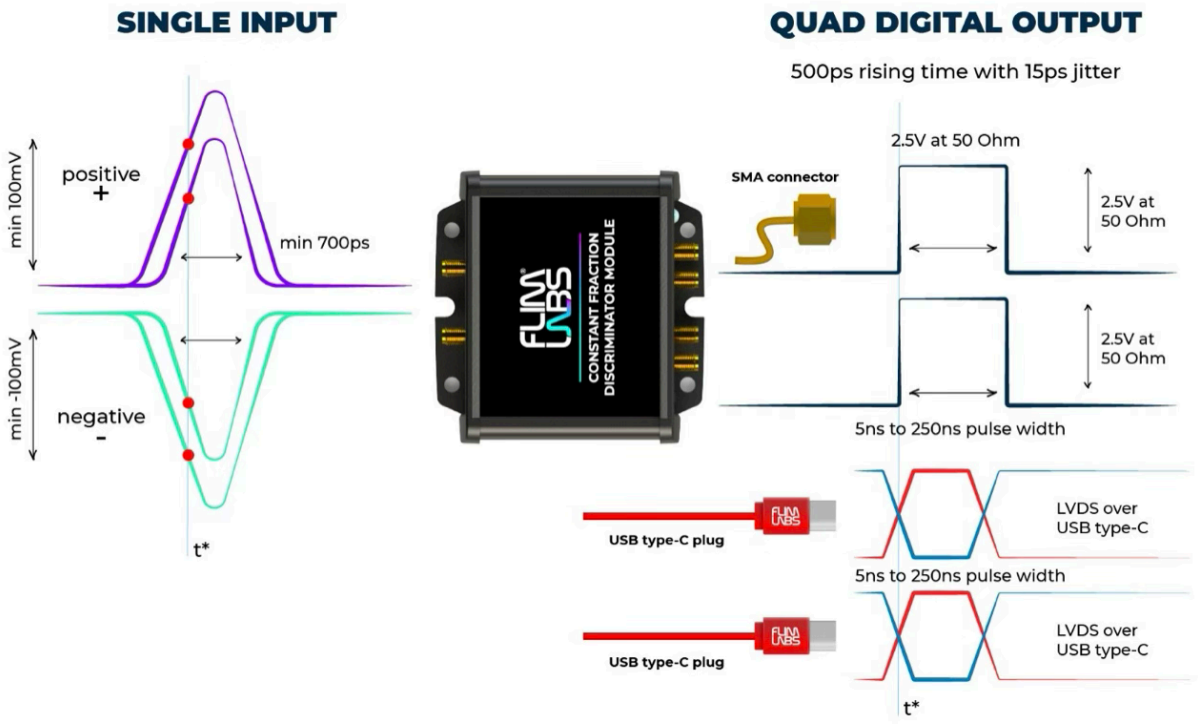


Figure 5.1: Input and output signal of a configured CFD.

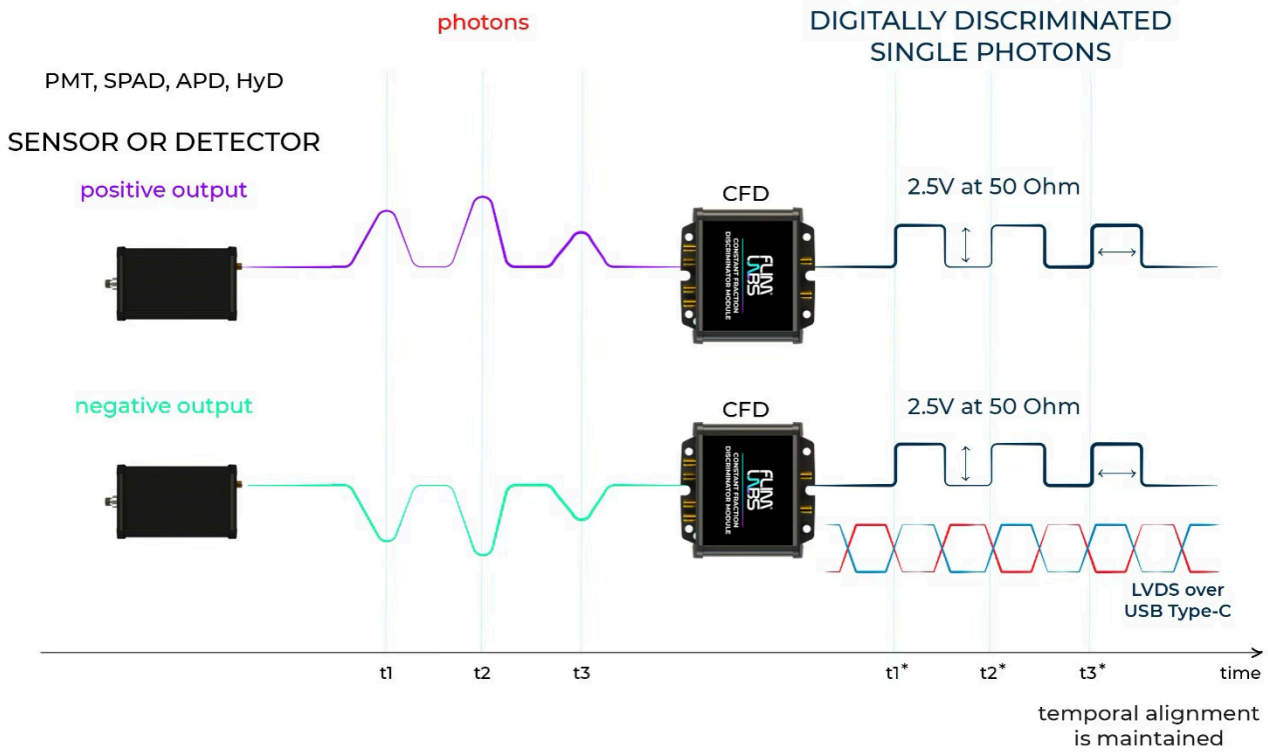


Figure 5.2: Illustration of the constant fraction discrimination process showing how positive and negative input pulses are converted into precise digital timing signals.

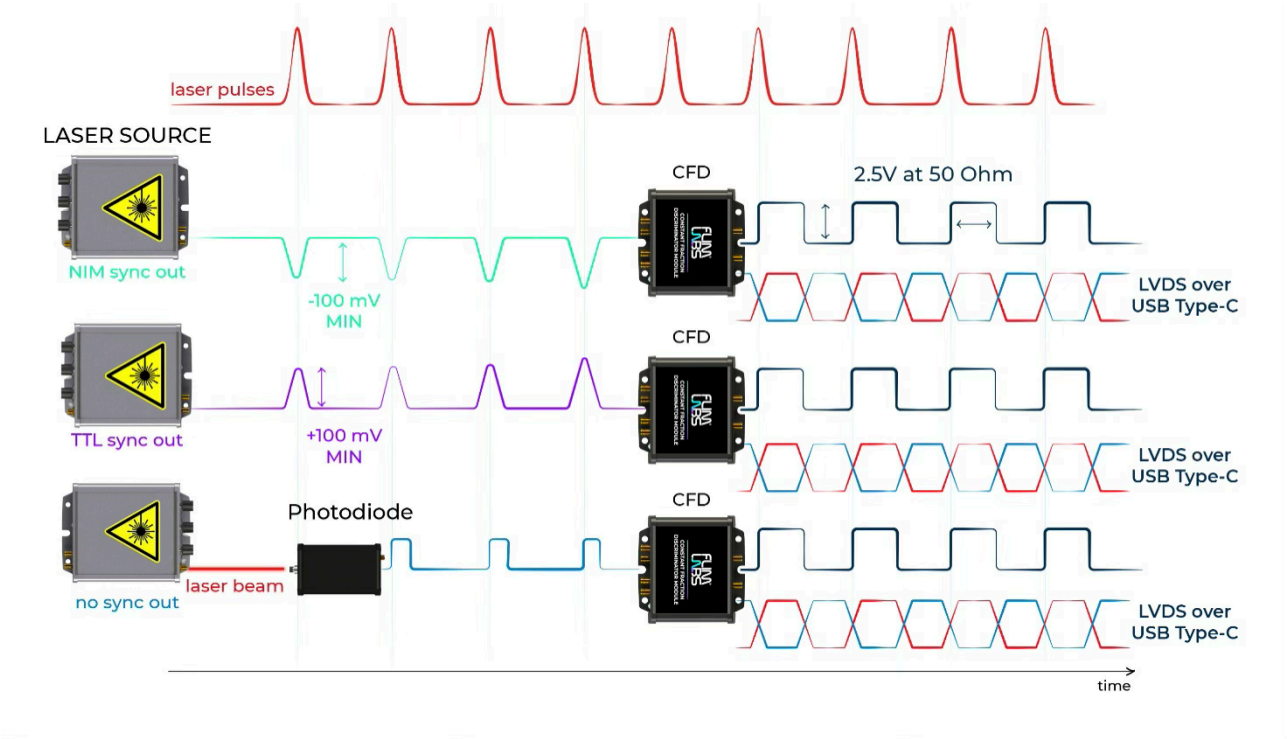


Figure 5.3: Illustration of various use cases of the Constant Fraction Discriminator.

5.4 Troubleshooting

Common faults and corrective actions are summarized in Table 5.1.

Table 5.1: Overview of common problems and recommended corrective actions.

Problem	Possible Cause	Recommended Action
Power and Device Status		
Device does not power on	Insufficient USB power	Use a USB port capable of delivering at least 900 mA at 5 V DC. Avoid passive USB hubs.
Device does not power on	Defective USB cable	Replace the USB cable and verify proper connector seating.
Input Signal		
No output signal	Input signal amplitude too low	Verify that the input signal amplitude exceeds the discriminator threshold.
No output signal	Incorrect polarity selection	Verify that the polarity selector matches the polarity of the input pulses.
No output signal	Improper impedance matching	Use 50 Ω coaxial cables and ensure proper termination.
Unstable triggering	Noise on input signal	Improve signal quality, reduce cable length, and ensure proper shielding and grounding.
CFD Configuration		

Problem	Possible Cause	Recommended Action
No output signal	Delay line length not suitable	Select a shorter delay line for fast-rising signals and a longer delay line for slower rise times.
Timing jitter higher than expected	Incorrect delay line selection	Adjust the delay line to match the rise time of the input signal.
Output frequency lower than input frequency	Output pulse width too long	Reduce the output pulse width to avoid pulse overlap at high count rates.
Output Signal		
No signal detected by acquisition system	Incorrect output connection	Verify that the SMA output is connected to the correct input channel of the acquisition system.
Output signal distorted	Incorrect cabling or excessive cable length	Use short 50 Ω coaxial cables and avoid adapters or T-connectors.



If any unexpected or potentially hazardous anomaly occurs during startup or operation, switch off the device immediately and disconnect it from the power source. Contact technical support before further use.

6 Maintenance and Service

This section describes the cleaning, preventive maintenance, corrective maintenance, and authorized service procedures for the device. All maintenance activities must be performed by qualified personnel who have read and understood this manual.



Before performing any cleaning or physical maintenance operation, the device must be switched off and disconnected from the power source.

Maintenance must be carried out in compliance with this manual and applicable workplace safety regulations. The maintenance area should provide adequate illumination; a minimum lighting level of 200 lux is recommended.

6.1 Cleaning

The device requires periodic external cleaning to ensure reliable operation.

- Remove dust or dirt from the exterior surfaces using a dry or slightly damp lint-free cloth.
- A mild, neutral detergent may be used if necessary.
- Do not allow liquids to enter connectors or internal components.
- Avoid the use of solvents, aggressive chemicals, or abrasive materials.

For connectors and hard-to-reach areas (e.g., USB ports), a soft-bristled brush or compressed air may be used.

6.2 Preventive Maintenance

The device does not require routine internal maintenance.
Periodically inspect:

- External surfaces for mechanical damage
- Connectors and cables for wear or deformation
- Product label for legibility. See Section 1.5 for label content and handling requirements.

If any abnormal condition is detected, discontinue use and contact technical support.

6.3 Service and Repairs

The device contains no user-serviceable internal components. In the event of malfunction, accidental damage, or improper operation:

- Switch off and disconnect the device.
- Do not attempt internal repair.
- Contact the manufacturer’s technical service for instructions.

Repairs or modifications must be carried out exclusively by FLIM LABS S.r.l. or authorized personnel. Use only original FLIM LABS spare parts. Further warranty conditions are described in Section 1.6.1.



Unauthorized intervention or the use of non-original components may compromise safety, performance, and warranty coverage.

7 Decommissioning and Disposal

This section describes procedures for removing the device from service and disposing of it in accordance with applicable regulations.

7.1 Decommissioning Procedure

The device is designed for long-term laboratory use. When it reaches the end of its technical or operational life, it must be decommissioned to prevent further use.

Decommissioning requires:

- Disconnecting the device from all power sources.
- Removing all signal connections.
- Ensuring that the device cannot be unintentionally reused.

The same procedure applies in cases of:

- Permanent withdrawal from service
- Storage without intended future use
- Final disposal

Reuse of components after final decommissioning is not permitted unless explicitly authorized by the manufacturer.

7.2 Disposal

The device must be disposed of in accordance with applicable local and European regulations governing electrical and electronic equipment. In accordance with Directive 2012/19/EU (WEEE), the device must not be disposed of with household waste. The product label includes the WEEE symbol indicating the requirement for separate collection of electrical and electronic equipment (see Section 1.5). Contact your local waste management authority or certified collection center for proper disposal procedures.

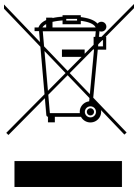


Figure 7.1: WEEE symbol